2ND SEMESTER LAB REPORT

TITLE : Electrical Circuit Analysis Lab



Faiq Ali (249520)

Muhammad Zeeshan (243346)

Muhammad Waqas (240756)

BACHELOR OF COMPUTER ENGINEERING (Fall-2025)

Submitted To:

Engr. Shanzay Jamil

Lab Engineer

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING FACULTY OF ENGINEERING AIR UNIVERSITY, ISLAMABAD

	AIR UNIVERSITY
	DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
	EXPERIMENT NO 14

Lab Title: PCB LAYOUT DESIGNING ON PROTEUS

Objectiv: PCB LAYOUT DESIGNING ON PROTEUS

Student Name: Muhammad Waqas -Muhammad Zeesha -Faiq Ali

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks:

Date: 8/May/2025 Signature:

OPENHANDED
LABORATORY
EXPERIMENT
NO. 14

PCB LAYOUT DESIGNING ON PROTEUS

OBJECTIVE:

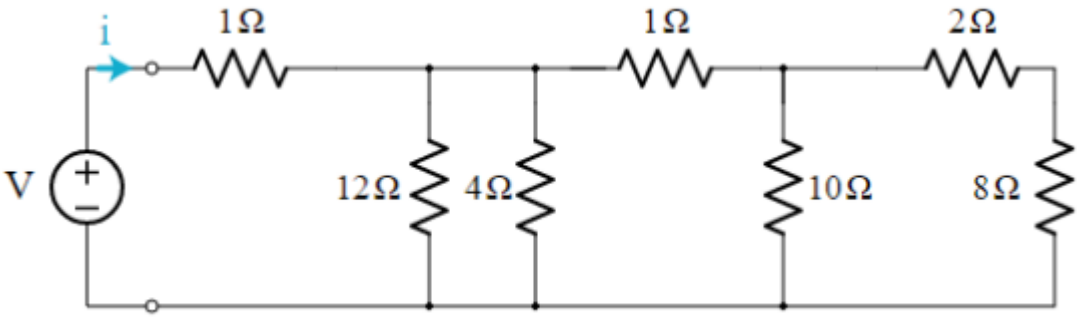
Build a complete and correct CB circuit.

1. Make sure voice messages send and receive.
2. Simulate channel changes and signal strength.
3. Learn how circuit parts affect communication.

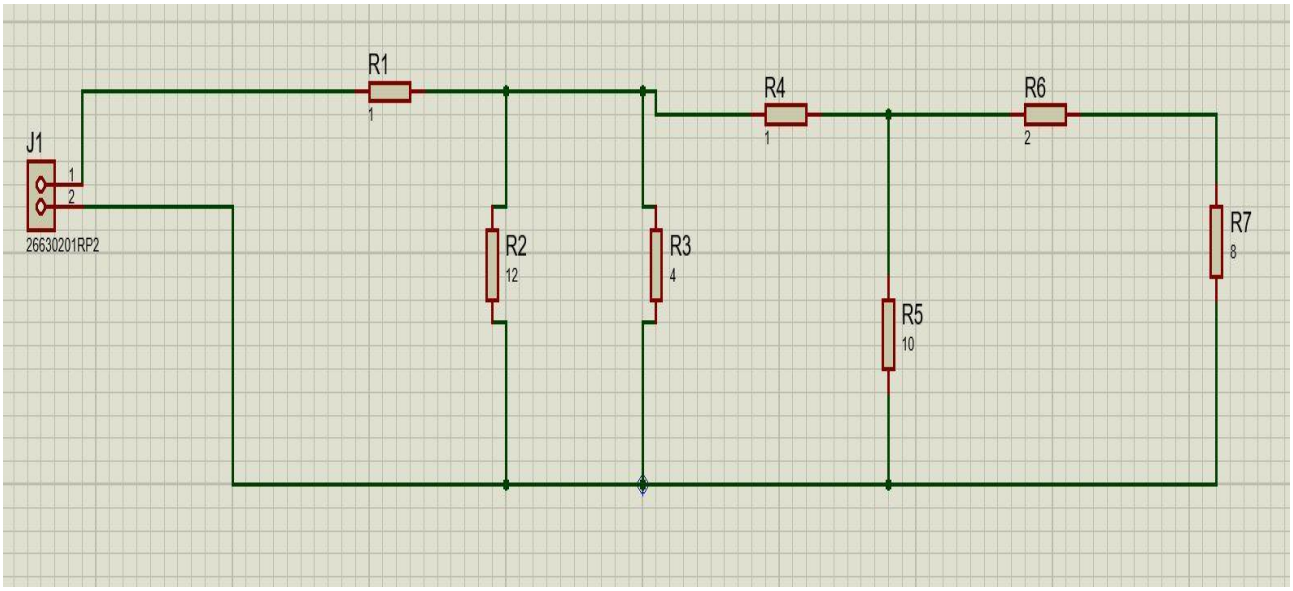
PCB LAYOUT DESIGNING

T PCB means Printed Circuit Board. This is a circuit board with printed copper layout connections. In dotted PCB board only dots are available. According to our requirement we can place or insert the components in those holes and attach the components with wires and soldering lid. In this dotted PCB we can make the circuit as out wish but it is very hard to design. There are so many difficulties are there. Proteus professional is a software combination of ISIS schematic capture program and ARES PCB layout program. This is a powerful and integrated development environment. Tools in this suit are very easy to use and these tools are very useful in education and professional PCB designing. As professional PCB designing software with integrated space based auto router, it provides features such as fully featured schematic capture, highly configurable design rules, interactive SPICE circuit simulator, extensive support for power planes, industry standard CADCAM & ODB++ output and integrated 3D viewer.

Lab task 1:



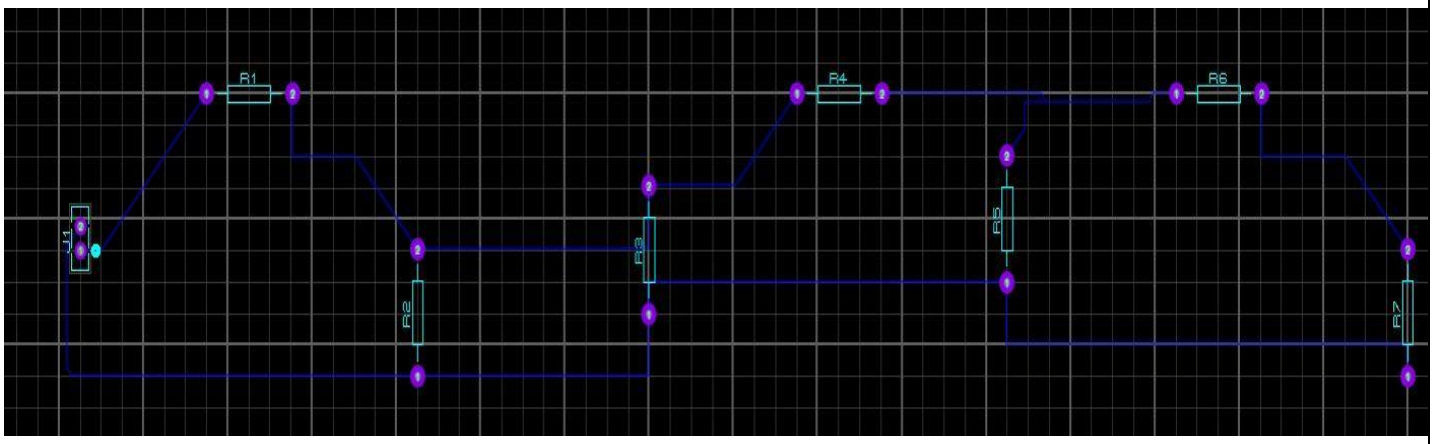
CIRCUIT DIAGRAM:



3D VISUALATION:






R1
R4
R6
J1
R2
R3
R5
R7

Conclusion:

In conclusion, this Proteus simulation successfully demonstrated the core functionalities of a CB radio system. By accurately building the circuit, we confirmed its ability to transmit and receive voice messages effectively. The simulation further allowed us to observe the practical implications of channel selection and varying signal strengths, providing valuable insight into how different components contribute to overall communication performance.